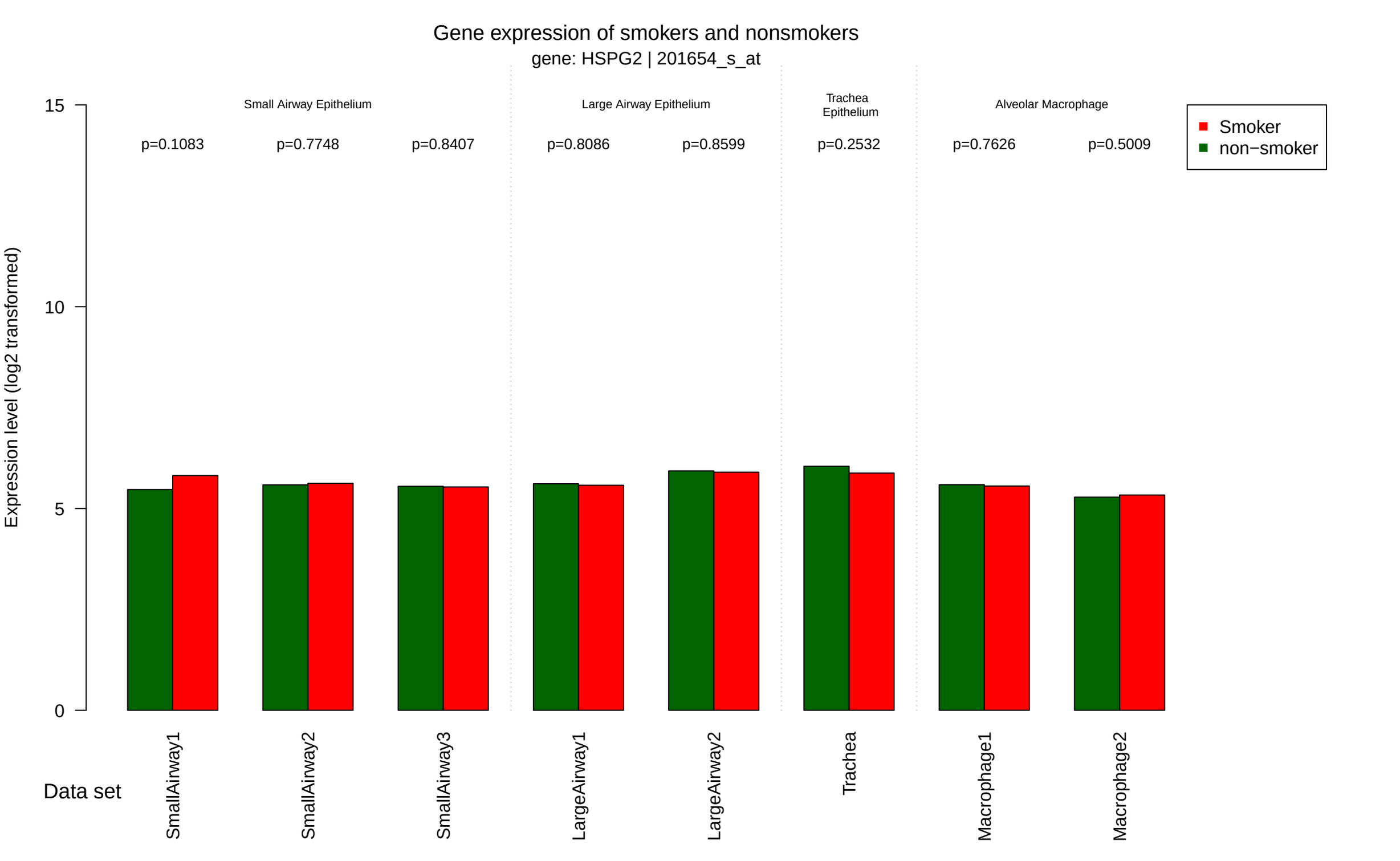
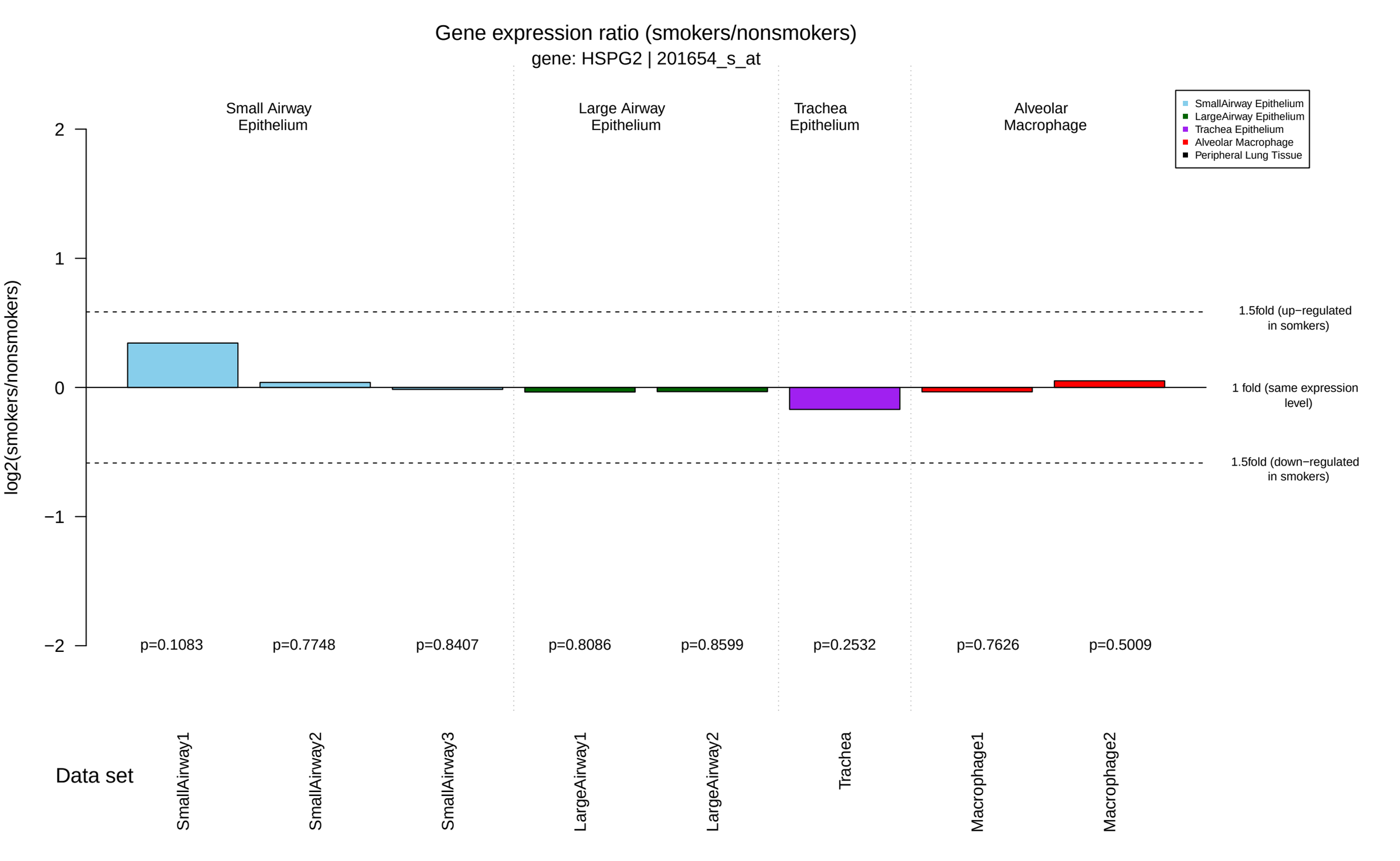
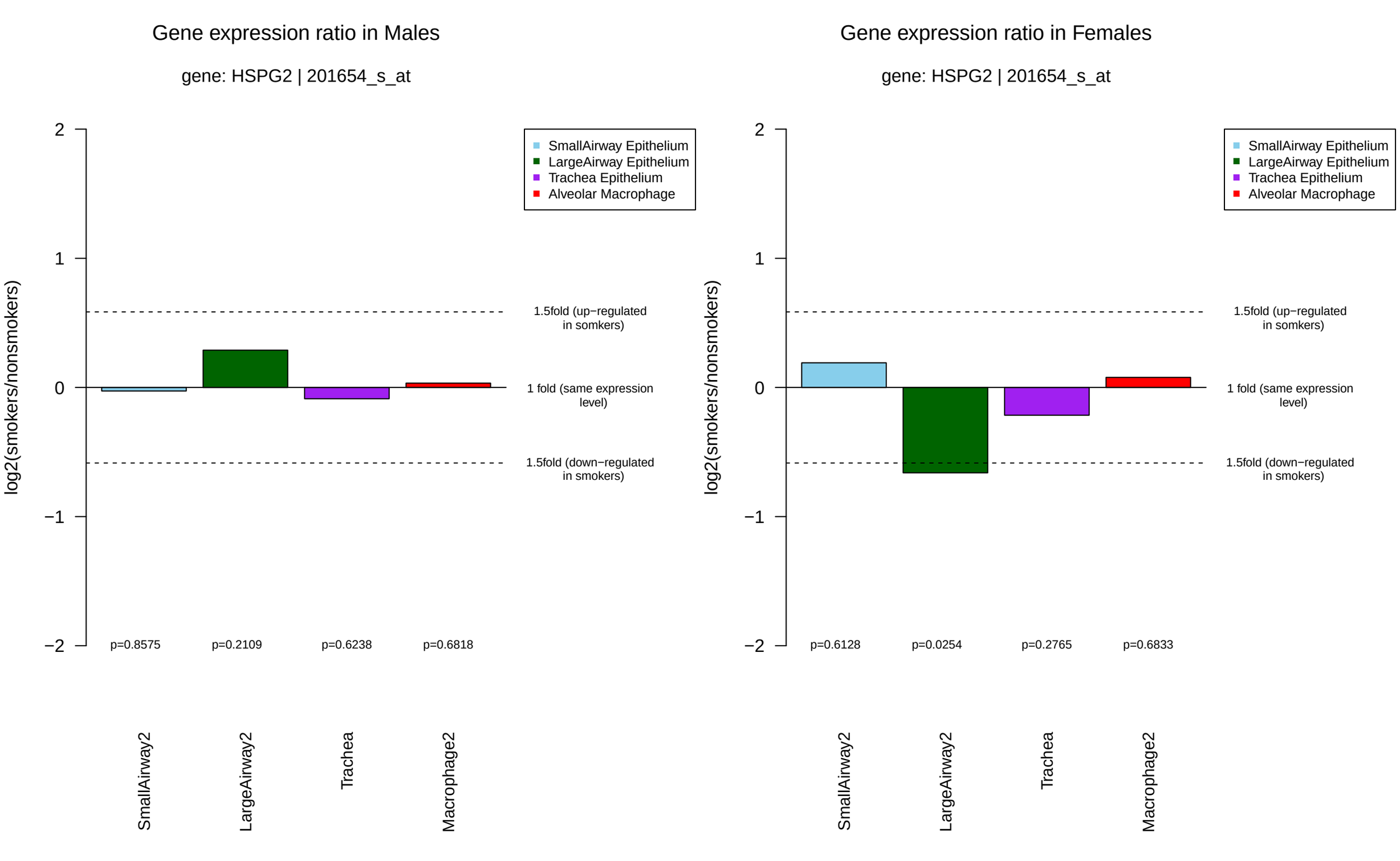
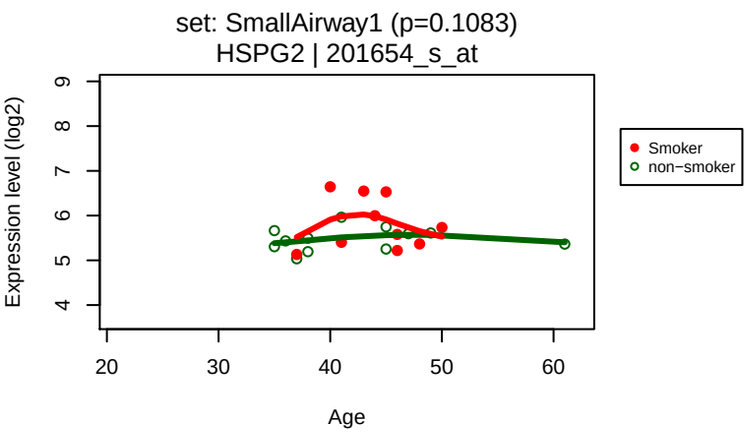
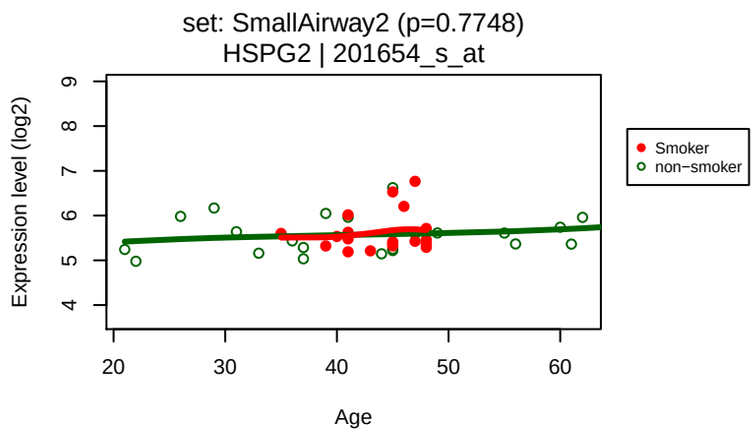
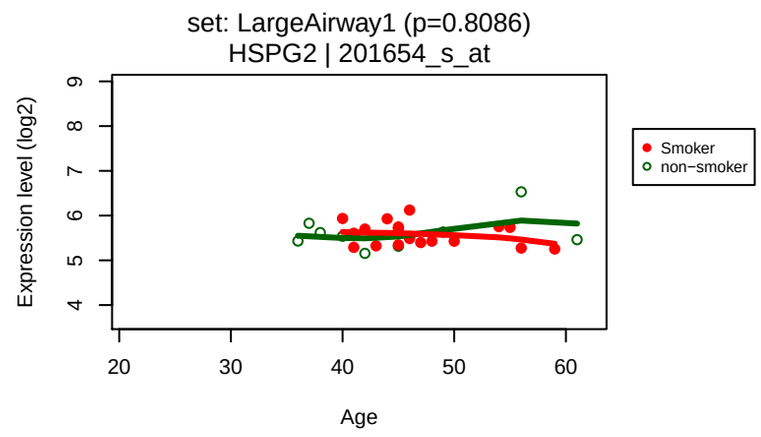

Gene expression of smokers and nonsmokers
gene: HSPG2 | 201654_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1083
p=0.7748
p=0.8407
p=0.8086
p=0.8599
p=0.2532
p=0.7626
p=0.5009
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

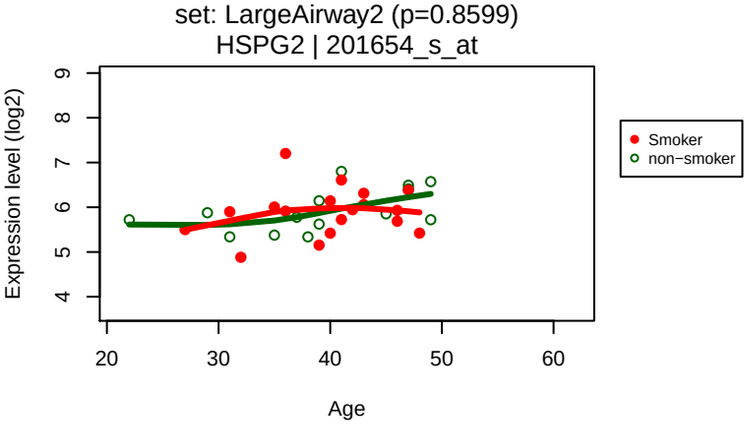

Gene expression ratio (smokers/nonsmokers)
gene: HSPG2 | 201654_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1083
p=0.7748
p=0.8407
p=0.8086
p=0.8599
p=0.2532
p=0.7626
p=0.5009
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

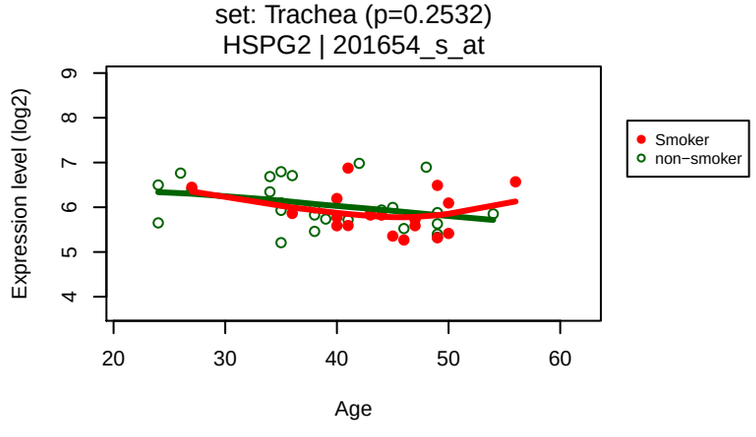

Gene expression ratio in Males
gene: HSPG2 | 201654_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8575
p=0.2109
p=0.6238
p=0.6818
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPG2 | 201654_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6128
p=0.0254
p=0.2765
p=0.6833
SmallAirway2
LargeAirway2
Trachea
Macrophage2

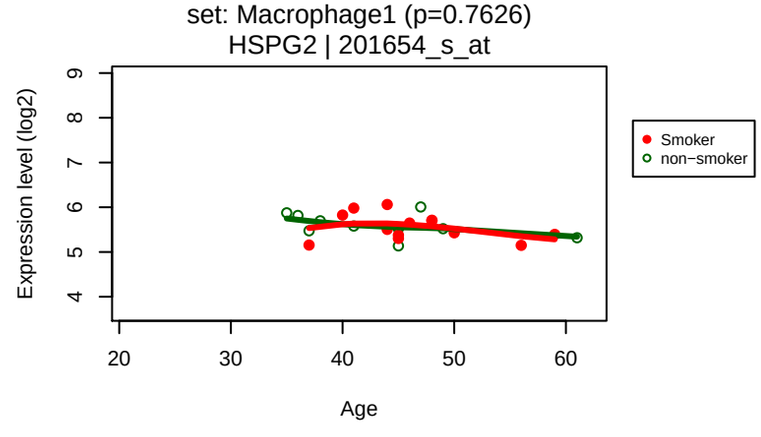

set: SmallAirway1 (p=0.1083)
HSPG2 | 201654_s_at
Expression level (log2)
Age
Smoker
non−smoker

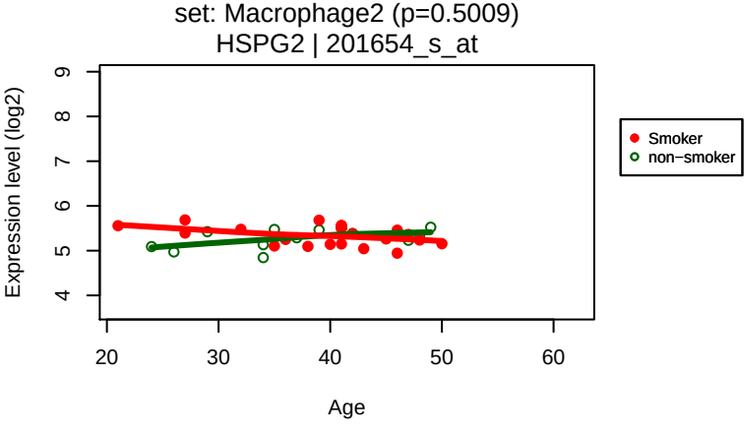

set: SmallAirway2 (p=0.7748)
HSPG2 | 201654_s_at
Expression level (log2)
Age
Smoker
non−smoker

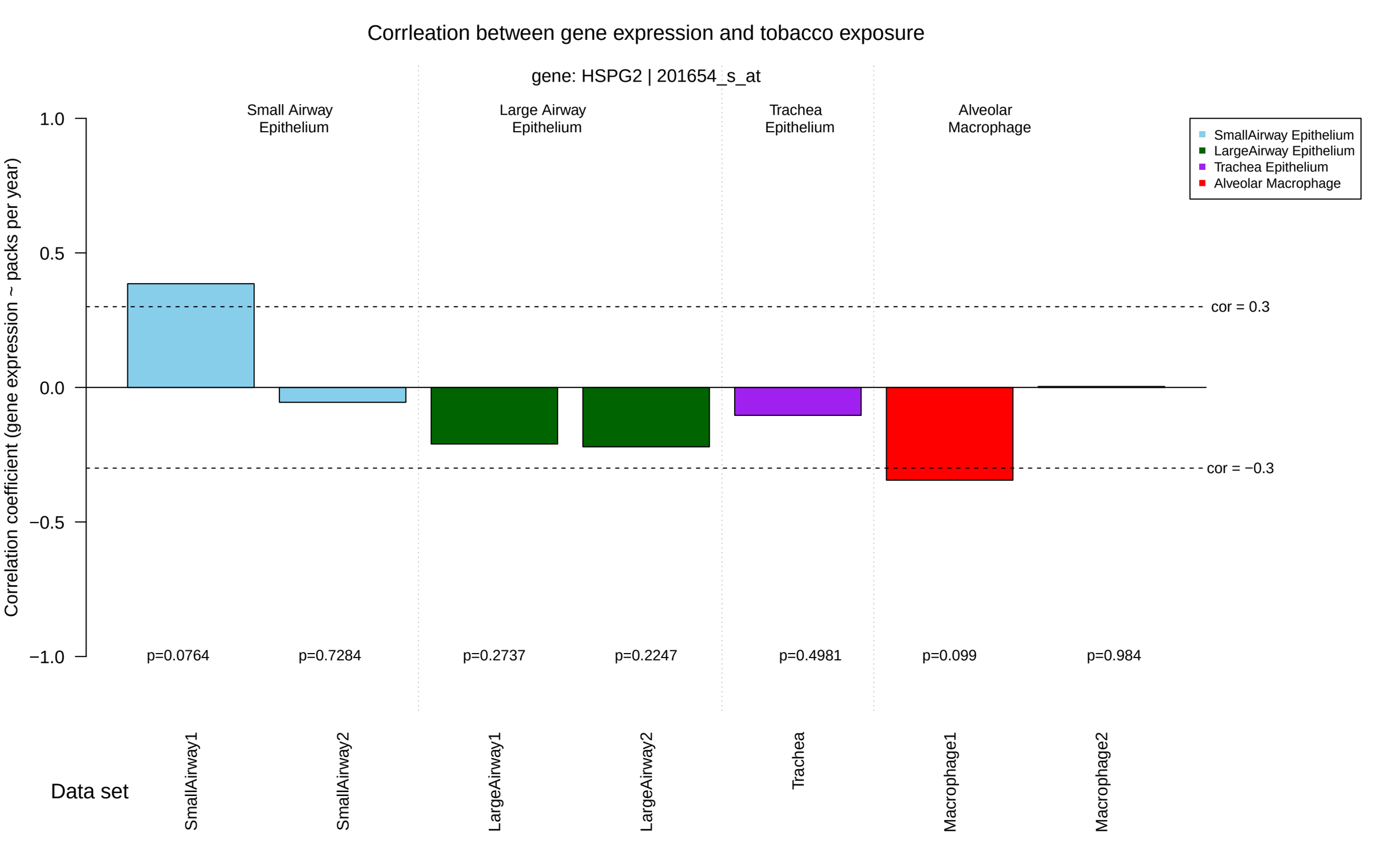

set: LargeAirway1 (p=0.8086)
HSPG2 | 201654_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8599)
HSPG2 | 201654_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2532)
HSPG2 | 201654_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7626)
HSPG2 | 201654_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5009)
HSPG2 | 201654_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPG2 | 201654_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0764
p=0.7284
p=0.2737
p=0.2247
p=0.4981
p=0.099
p=0.984
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2